Phase 3 – Work Package 1: BRFSS Data Provenance and Dataset Integrity Audit

Project: Cardiovascular Disease Prediction Using Machine Learning: A BRFSS-Based Study with Temporal Validation

Deliverable ID: PHASE-03-WP1

Document type: Technical Audit Record

Document version: 1.0.0

Audit date: 1 November 2026

Work package status: Completed with documented limitations

Working branch: `phase/03-data-provenance-audit`

1. Purpose and position within Phase 3

This document records the data provenance, dataset integrity, schema compatibility and outcome-consistency audit conducted for the BRFSS datasets used in the cardiovascular disease prediction research.

The audit is a supporting technical deliverable within Phase 3. It establishes a traceable record of the source datasets and verifies that the locally stored outcome variable is consistent with the documented CDC calculation rule.

It is not the final report for Phase 3.

Phase 3 contains separate research activities:

Work package	Description	Status
WP1	BRFSS data provenance and dataset integrity audit	Completed with documented limitations
WP2	Scientific manuscript development	Separate ongoing activity

The completion of this audit does not indicate that manuscript development, journal selection or submission preparation has been completed.

1.1 Audit objectives

The audit was conducted to:

1. Record the provenance and integrity of the local BRFSS source files.
2. Calculate and preserve SHA-256 checksums for the inspected datasets.
3. Inspect respondent counts and variable schemas.
4. Verify the availability of the six locked model predictors.
5. Identify annual differences affecting dataset compatibility.
6. Verify the local `_MICHD` outcome against the CDC calculation rule.
7. Profile missingness in the outcome and core predictors.
8. Preserve the original Phase 1 model specification and historical results.
9. Produce reproducible audit artefacts for subsequent manuscript development.

1.2 Audit boundaries

The audit was designed as a read-only verification activity.

It did not:

- Retrain or refit the primary Gradient Boosting model.
- Change the six locked predictors.
- Change the outcome definition.
- Modify the historical train, validation or test results.
- Introduce substitute predictors for annual compatibility.
- Replace the Phase 1 computational baseline.
- Claim clinical validation or prospective predictive performance.

2. Audit outcome at a glance

Overall audit outcome: `PASS_WITH_DOCUMENTED_LIMITATIONS`

Audit area	Finding	Outcome
Local source-file integrity	File sizes and SHA-256 checksums recorded	Passed
Dataset respondent counts	Local counts recorded for all three years	Passed
Local schema inspection	No duplicate variable names identified	Passed
Predictor availability	All six predictors available in 2023 and 2025	Passed
2024 compatibility	`BPHIGH6` unavailable	Compatibility audit only
`_MICHD` target consistency	Zero mismatches across 1,247,151 records	Passed
Missingness profiling	Core-variable profile generated	Completed
CDC variable-count reconciliation	Differences remain across official documentation and local files	Open limitation
Phase 1 model preservation	No retraining or modification performed	Preserved

The audit successfully verifies the recorded local dataset properties and `_MICHD` calculation consistency. However, differences in the variable counts reported by CDC documentation and the inspected local files remain documented and require source-version reconciliation before the manuscript is finalised.

3. Dataset provenance and file integrity

Three annual BRFSS combined landline and cellular telephone datasets were inspected.

3.1 Local source files

Survey year	Repository-relative source file	File size (bytes)	SHA-256
2023	`data/raw/brfss/2023/LLCP2023.XPT`	1,205,554,400	`3d3bf8ef5195bde227828ddc4c90745b76e8b304f8f5b9a043b6d99895fd1615`
2024	`data/raw/brfss/2024/LLCP2024.XPT`	1,093,874,240	`7d53cf088f858e02bcc029a51cf0408cc68f40a0ebeb02a3c0f73058aab36b1c`

Survey year	Repository-relative source file	File size (bytes)	SHA-256
2025	`data/raw/brfss/2025/LLCP2025.XPT`	799,971,280	`d99262f8854f018a600c3538ee78e4308569af700abe6b92b36180336c24dec2`

The SHA-256 values provide local integrity identifiers for the inspected files. They can be used to determine whether the files change during subsequent research activities.

These are locally calculated checksums, not CDC-published checksums.

3.2 Raw-data version-control policy

The raw BRFSS XPT files are excluded from Git because of their large file sizes.

The repository retains:

- Dataset provenance metadata.
- File sizes and SHA-256 checksums.
- Audit scripts.
- Machine-readable audit results.
- Relevant archived CDC documentation.

The original raw datasets remain local research inputs and are not committed to the GitHub repository.

4. Dataset structure and annual compatibility

4.1 Local schema inspection

The local XPT files were inspected for respondent counts, variable counts, duplicate variable names and the availability of the required research variables.

Survey year	Respondents	Variables in local XPT	Required predictors available	Local audit interpretation
2023	433,323	350	6 of 6	Suitable for model development
2024	457,670	301	5 of 6	Compatibility audit only
2025	356,158	283	6 of 6	Suitable for temporal evaluation

No duplicate variable names were identified in the inspected local schemas.

4.2 Predictor availability

The 2023 and 2025 local datasets contain all six locked predictors.

The 2024 local dataset does not contain `BPHIGH6`, the blood-pressure predictor used in the locked model specification.

Consequently:

- 2024 is retained for compatibility and variable-availability assessment only.
- It is not used as a replacement development or temporal evaluation dataset.
- No alternative blood-pressure variable is introduced.
- The six-predictor model specification remains unchanged.

4.3 CDC variable-count reconciliation

A source-documentation discrepancy was identified when comparing the local XPT files with CDC's published annual documentation.

Survey year	Local XPT count	CDC annual data page	CDC cross-year summary table
2023	350	345	350
2024	301	345	301
2025	283	284	284

The CDC annual data pages and the CDC cross-year summary table do not report identical counts for every year.

For 2023 and 2024, the local XPT counts agree with the CDC cross-year summary table, but differ from the counts stated on the respective annual data pages.

For 2025, the local XPT contains 283 variables. The archived detailed CDC variable layout also identifies 283 variables, and the local XPT and archived layout have an exact variable-name match. However, the CDC annual data page and cross-year summary table report 284 variables.

These differences are recorded as unresolved documentation and source-version discrepancies.

They must not be silently corrected or interpreted as proof that the local files are identical to every currently published CDC data package.

4.4 Source-version verification still required

Before finalising the manuscript's data-source description, the following should be reconciled:

1. The exact CDC release or file version represented by each locally stored XPT file.
2. The difference between the 2023 and 2024 annual-page variable counts and the cross-year summary counts.
3. The one-variable difference between the 2025 local XPT/detailed layout and the CDC annual summary.
4. The variable-name lists associated with the relevant CDC releases.

The existing local audit results are retained as observations of the files actually inspected.

5. Locked outcome and predictor definitions

The audit preserved the outcome and predictor definitions used in the original Phase 1 experiment.

5.1 Outcome variable

The target variable is:

`_MICHD`

It represents prevalent self-reported history of myocardial infarction or coronary heart disease.

The source variables used in its derivation are:

- `CVDINFR4`
- `CVDCRHD4`

These source variables are used to calculate the outcome and must not be included as model predictors.

5.2 Locked predictor set

BRFSS source variable	Research variable	Role
`_AGEG5YR`	`AGE_GROUP`	Age group
`EXERANY2`	`PHYSICAL_ACTIVITY`	Physical activity
`_SMOKER3`	`SMOKING_STATUS`	Smoking status
`BPHIGH6`	`BLOOD_PRESSURE`	Blood-pressure status
`DIABETE4`	`DIABETES_STATUS`	Diabetes status
`_BMI5`	`BMI`	Body mass index

The six predictors remain fixed.

5.3 Excluded variables

The following variables are explicitly excluded from the ordinary model predictor set:

Variable	Reason for exclusion
`_MICHD`	Target variable
`CVDINFR4`	Source variable used to derive the target
`CVDCRHD4`	Source variable used to derive the target
`_LLCPWT`	Survey weight
`_STSTR`	Survey stratum
`_PSU`	Primary sampling unit

Survey-design variables may be retained for appropriately labelled survey-weighted analyses. They are not treated as ordinary model predictors.

6. `_MICHD` target-consistency verification

6.1 CDC calculation rule

The CDC calculated-variable documentation defines `_MICHD` using `CVDINFR4` and `CVDCRHD4`.

The documented logic is:

- `_MICHD = 1` when either source variable indicates a reported heart attack or angina/coronary heart disease.
- `_MICHD = 2` when both source variables indicate that the respondent did not report either condition.
- Otherwise, `_MICHD` is missing.

The audit reapplied this rule to the local annual datasets and compared the expected values with the stored `_MICHD` values.

6.2 Verification results

Survey year	Records checked	Expected defined outcomes	Expected missing outcomes	Observed missing outcomes	Mismatches
2023	433,323	428,738	4,585	4,585	0
2024	457,670	452,464	5,206	5,206	0
2025	356,158	352,145	4,013	4,013	0
Total	**1,247,151**	**1,233,347**	**13,804**	**13,804**	**0**

Result: Zero mismatches across 1,247,151 inspected records.

The stored `_MICHD` values agree with the documented CDC derivation rule for all three local datasets.

6.3 Interpretation

This verification establishes consistency between the locally stored outcome and the documented CDC calculation rule.

It does not:

- Independently verify the truth of respondents' original survey answers.
- Establish clinical diagnostic accuracy.
- Validate the outcome against medical records.
- Establish prospective disease risk.

The target remains a survey-derived indicator of self-reported prevalent disease history.

7. Core-variable profile and missingness

A core-variable profile was generated for the outcome, its two source variables and the six locked predictors.

The profile records variable labels and missing-value counts for each survey year.

7.1 Key missingness observations

Variable	2023 missing values	2024 missing values	2025 missing values
`_MICHD`	4,585	5,206	4,013
`_BMI5`	40,535	43,037	32,562
`BPHIGH6`	Available	Not present in local schema	2,617

The main observations are:

1. `_BMI5` has the highest missingness among the six predictors in each inspected annual dataset.
2. `BPHIGH6` is absent from the local 2024 dataset.
3. `BPHIGH6` has 2,617 missing values in the local 2025 dataset.
4. The missing outcome counts agree with the target-consistency audit.
5. Other core predictors have comparatively low missingness in the inspected files.

7.2 Missing predictors in the evaluation samples

The existing evaluation evidence also identifies records with at least one missing predictor.

Evaluation sample	Records with at least one missing predictor	Total evaluation records	Proportion
2023 held-out	8,200	64,311	12.75%
2025 temporal	47,595	352,145	13.52%

Performance differed between records with complete predictors and records with at least one missing predictor.

This should be considered during manuscript reporting. Missingness should not be treated as inconsequential simply because the existing preprocessing pipeline can process incomplete records.

The detailed profile is retained in:

```
artifacts/provenance/phase3_core_variable_profile.csv
```

8. Relationship to the locked Phase 1 experiment

The provenance audit did not retrain or modify the primary Gradient Boosting model.

The following Phase 1 decisions remain unchanged:

- Development dataset: BRFSS 2023.
- Temporal evaluation dataset: BRFSS 2025.
- Secondary compatibility dataset: BRFSS 2024.
- Predictor set: six locked predictors.
- Primary model: Gradient Boosting.
- Secondary model: L2 Logistic Regression.
- Existing preprocessing pipeline.
- Existing classification threshold.
- Historical train, validation, test and temporal evaluation results.

The 2025 dataset remains a temporal evaluation sample. It was not used to refit the model or select a new threshold.

8.1 Interpretation safeguards

The audit does not establish that the model is clinically validated or suitable for deployment.

The following limitations remain relevant to the manuscript:

- BRFSS is based on self-reported survey information.
- The datasets are cross-sectional annual survey samples.
- Missingness may affect observed performance.
- Survey-weighted point estimates are not equivalent to full design-based inference.
- Confidence intervals and full complex-survey variance estimation have not been established for the retained weighted results.
- Subgroup performance varies and should not be represented as uniform.

The audit therefore supports data-provenance transparency, not a claim of clinical validity.

9. Audit implementation and reproducibility

9.1 Audit script

The audit was implemented through:

`scripts/audit_brfss_provenance.py`

The script performs read-only checks of the local BRFSS files and supports an optional profile mode.

The profile mode generates the core-variable profile and target-consistency results.

9.2 Reproducibility command

From the dedicated audit Python environment containing `pandas` and `pyreadstat`, run:

```
python scripts/audit_brfss_provenance.py --profile
```